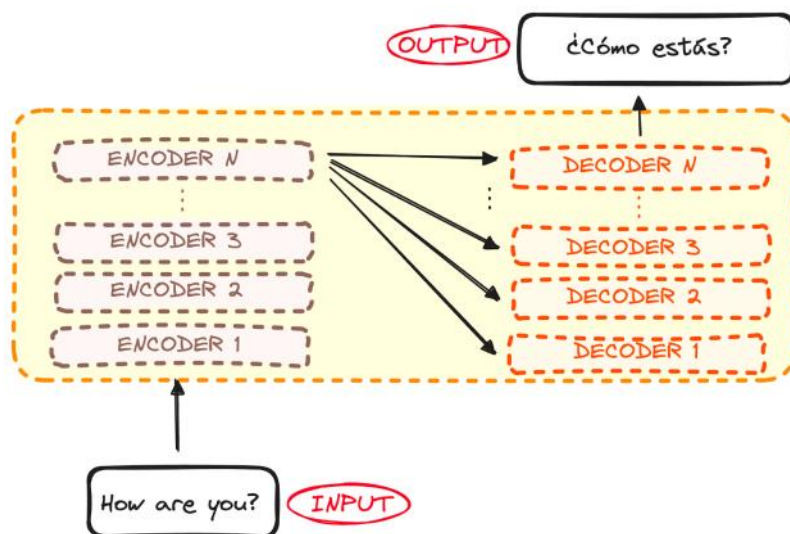
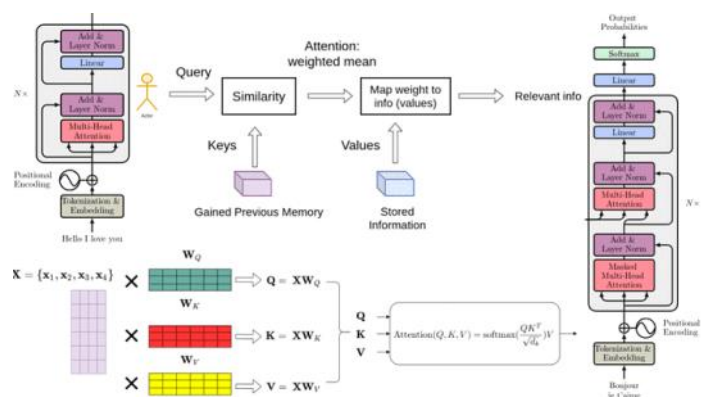
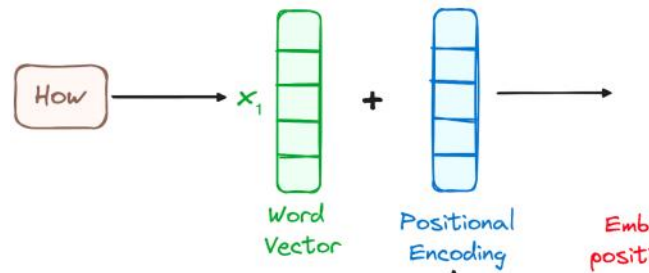
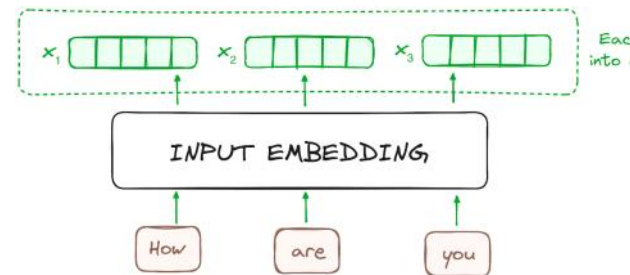
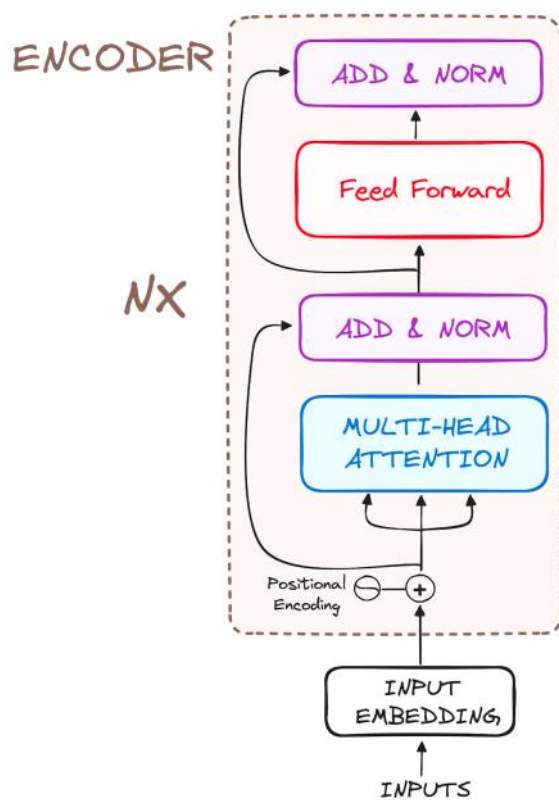
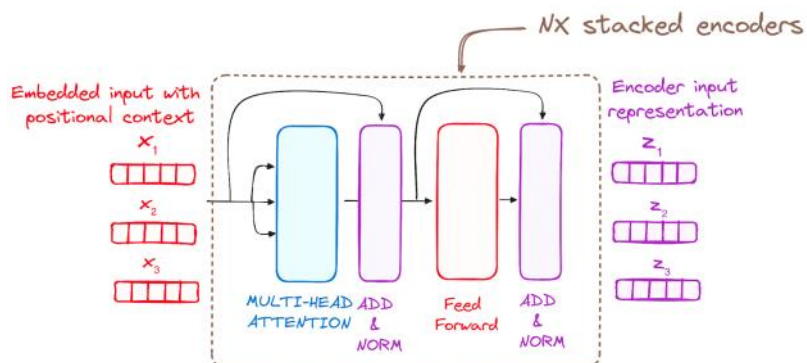
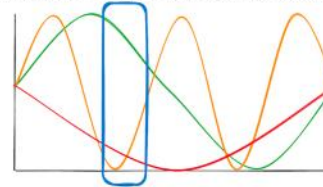






Binary encoding of sinus and cosinus determine if words are close to each



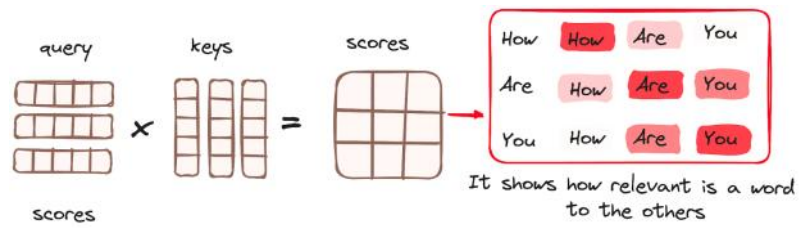
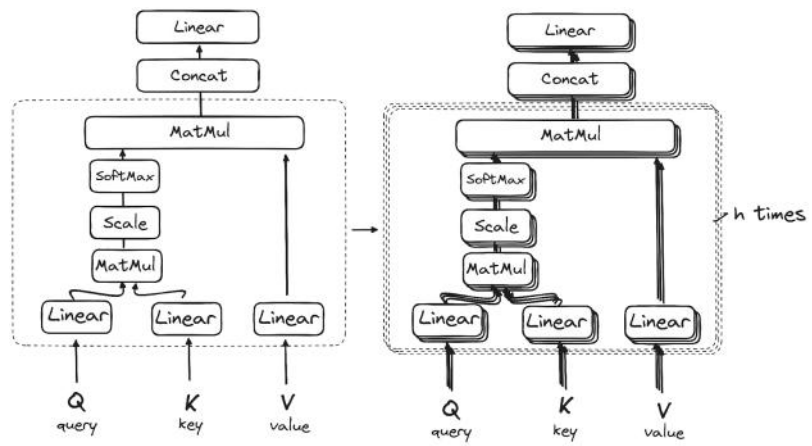
h word is embedded
a vector of size 512.

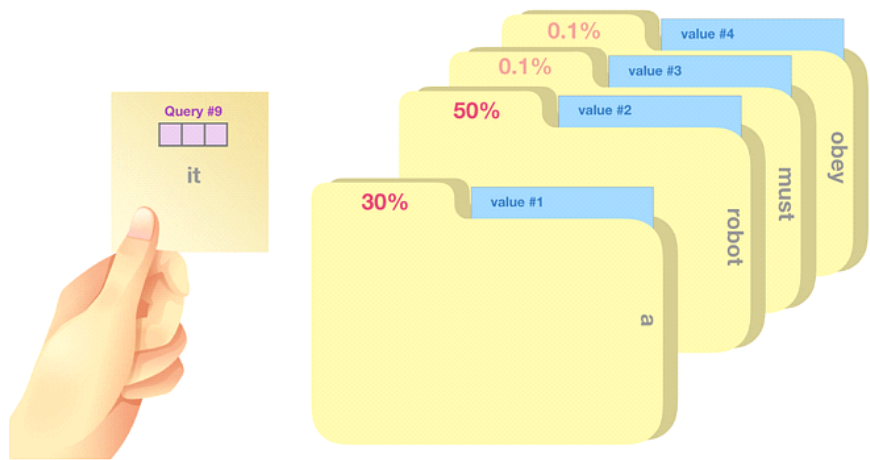


embedding with
positional context

to
other.



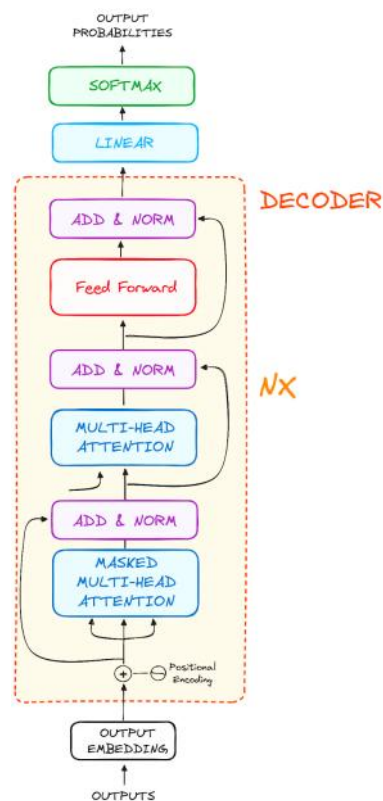


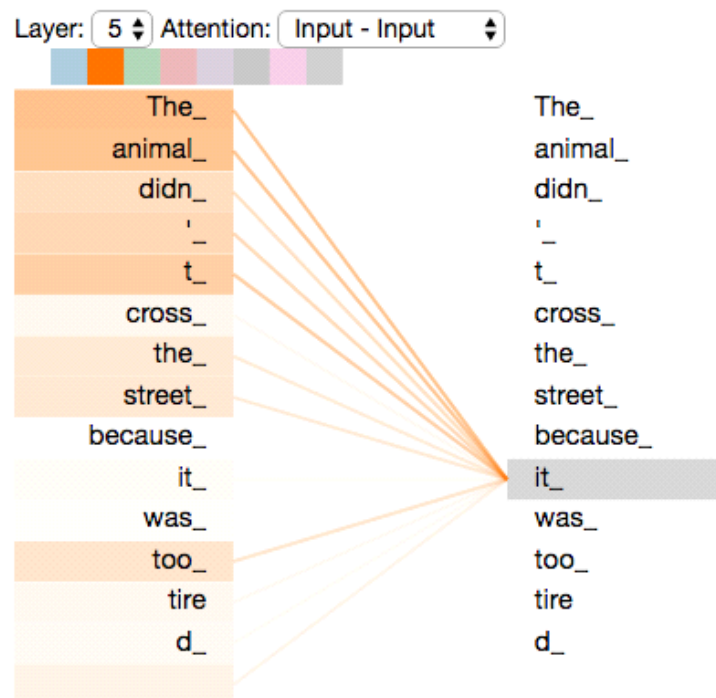
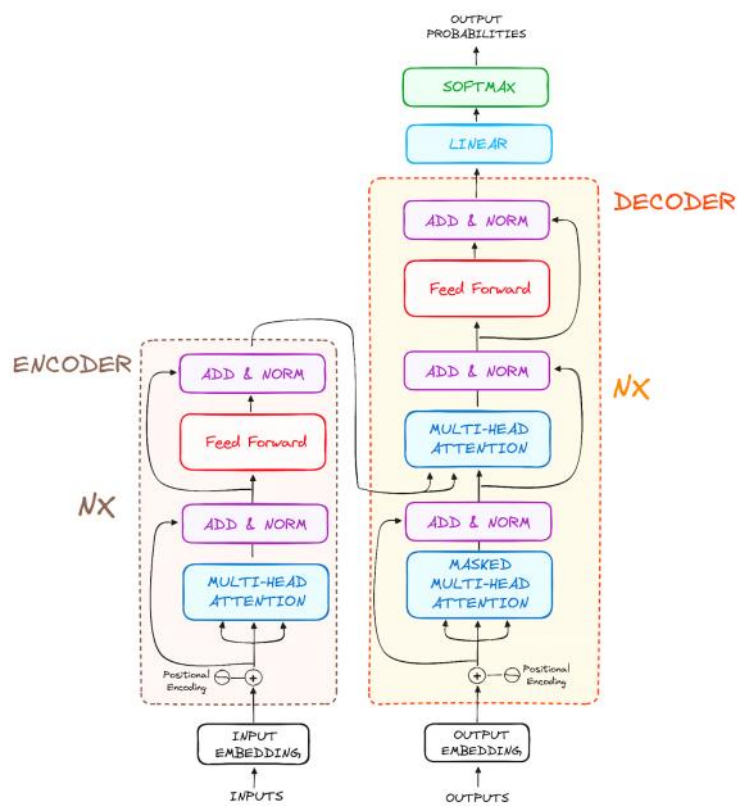


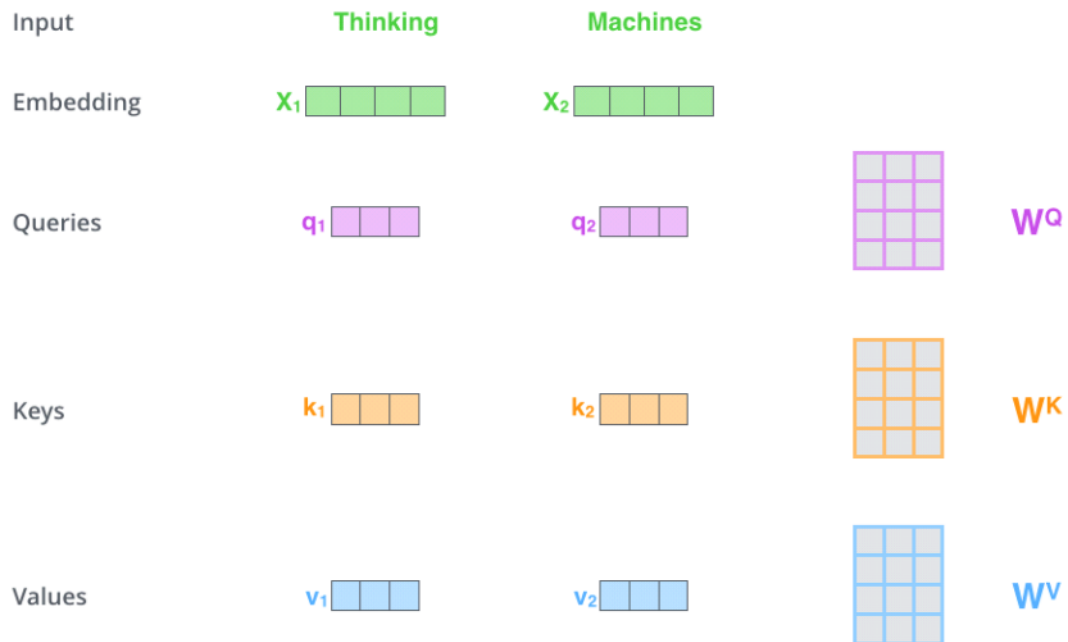
$$\text{Attention}(Q, K, V) = \frac{(Q \cdot K^T)}{\sqrt{d \times V}}$$

Higher scores get heighten, and lower scores are depressed

$$\text{Softmax}(\text{grid}) = \text{grid}$$







$$\begin{array}{c} X \\ \text{1x4 green grid} \end{array} \times \begin{array}{c} W^Q \\ \text{4x3 purple grid} \end{array} = \begin{array}{c} Q \\ \text{4x3 purple grid} \end{array}$$

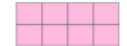
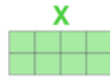
$$\begin{array}{c} X \\ \text{1x4 green grid} \end{array} \times \begin{array}{c} W^K \\ \text{4x3 orange grid} \end{array} = \begin{array}{c} K \\ \text{4x3 orange grid} \end{array}$$

$$\begin{array}{c} X \\ \text{1x4 green grid} \end{array} \times \begin{array}{c} W^V \\ \text{4x3 blue grid} \end{array} = \begin{array}{c} V \\ \text{4x3 blue grid} \end{array}$$

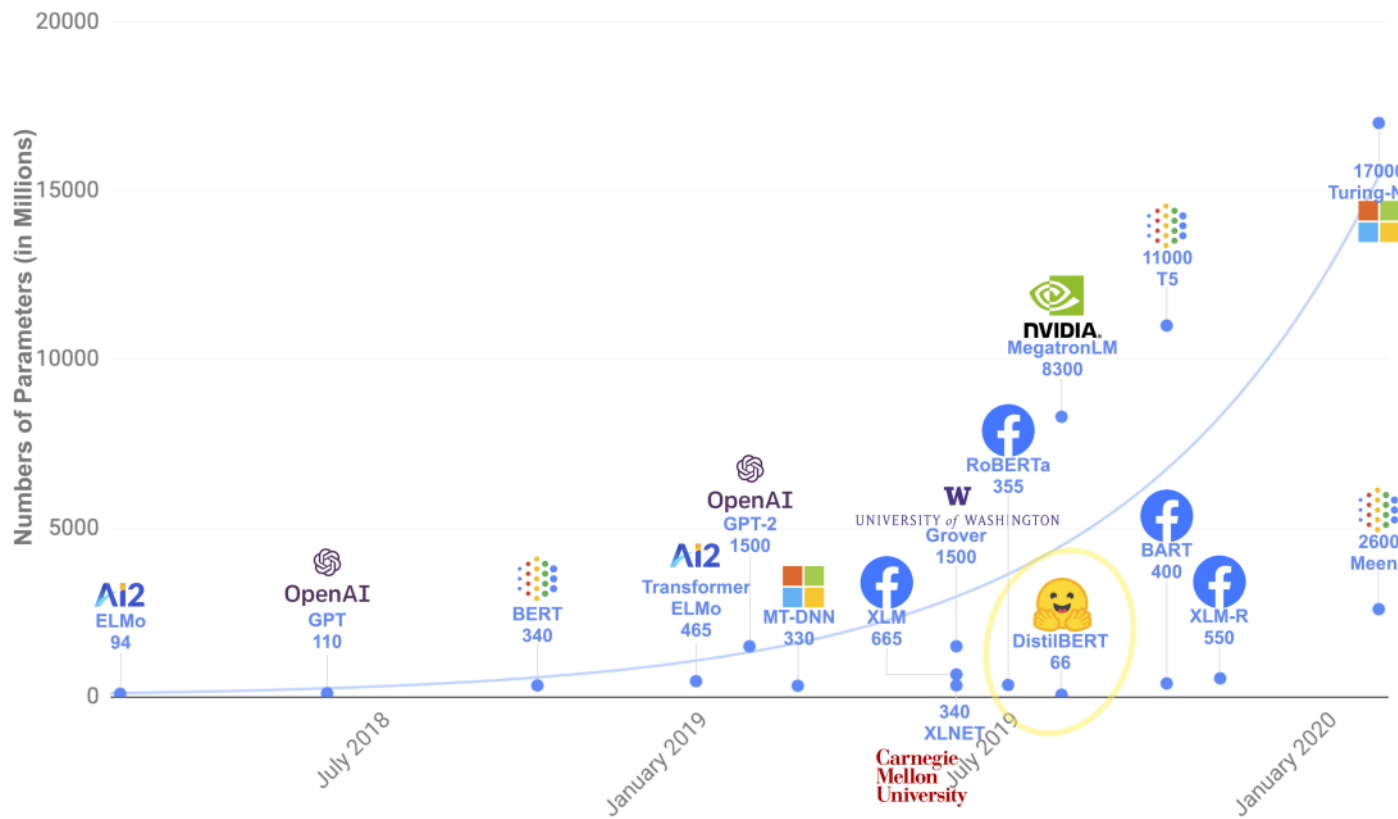
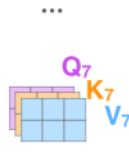
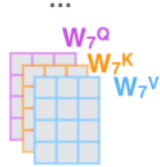
$$\begin{aligned}
 & \text{softmax} \left(\frac{\begin{array}{c} Q \\ \text{4x3 purple grid} \end{array} \times \begin{array}{c} K^T \\ \text{3x4 orange grid} \end{array}}{\sqrt{d_k}} \right) \begin{array}{c} V \\ \text{4x3 blue grid} \end{array} \\
 &= \begin{array}{c} Z \\ \text{4x3 pink grid} \end{array}
 \end{aligned}$$

- 1) This is our input sentence*
- 2) We embed each word*
- 3) Split into 8 heads. We multiply X or R with weight matrices
- 4) Calculate attention using the resulting $Q/K/V$ matrices
- 5) Concatenate the resulting Z matrices, then multiply with weight matrix W^O to produce the output of the layer

Thinking
Machines



* In all encoders other than #0, we don't need embedding. We start directly with the output of the encoder right below this one



The boy is holding a blue ball

0
NLG

a

The cat drank the milk because **it** was hungry
 The cat drank the milk ~~because~~ **it** was sweet

The
 cat
 drank
 the
 milk
 because
 it
 was
 hungry

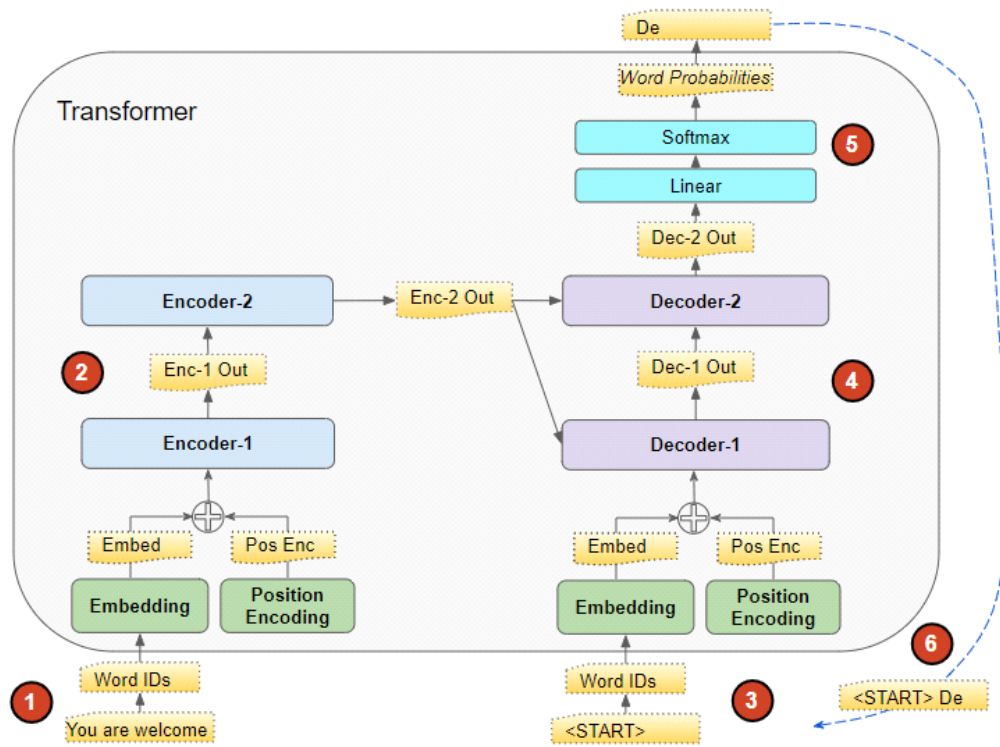
The
 cat
 drank
 the
 milk
 because
 it
 was
 hungry

The
 cat
 drank
 the
 milk
 because
 it
 was
 sweet

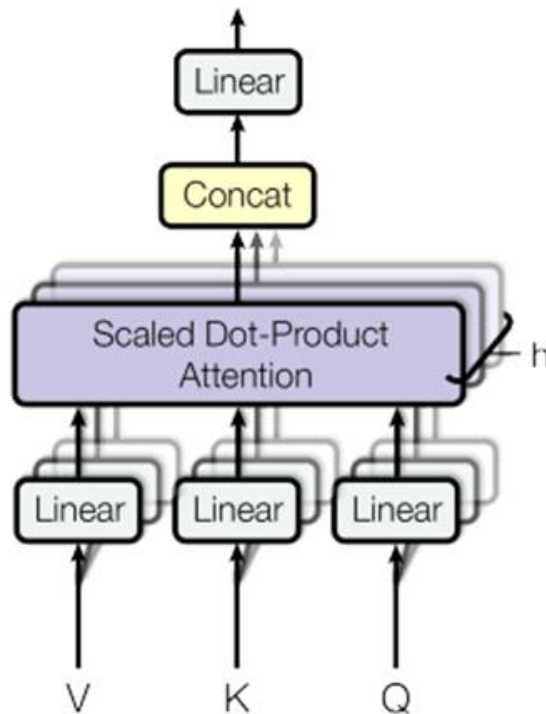
The
 cat
 drank
 the
 milk
 because
 it
 was
 sweet

g 7

k
use
t



Multi-Head Attention



Quadratic

→ O(n²)

the cat sat on the mat

sin

Con

(n^2)

mat the on sat ~~cat~~ the

